

README — novus-aexenti/

Cognitive Reasoning Engine & Asymmetric Dual-Brain MoE

W5+H Subsystem Identity

- **WHO:** Houses the 0.8B Triage / Intent Model and the 4B/9B Core Executor Model.
 - **WHAT:** Dual-agent routing logic, in-session `mem0` episodic state, and the `Reasoning Bank` RLVR recovery ledger.
 - **WHEN:** Fired on every incoming user prompt, voice transcription, or background task dispatch.
 - **WHERE:** `novae-xorpus/novus-aexenti/` (Federated Subsystem Root).
 - **WHY:** Solves the context bloat and RAM saturation problem by separating fast intent parsing from heavy code generation, keeping mobile RAM usage under 5.0 GB.
 - **HOW:** Dispatches small prompts to the 0.8B triage model, pulls habits from `mem0`, compiles clean meta-prompts, and logs step progress to `active_execution_paths.json`.
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Internal Directory Topology

novus-aexenti/

— README.md	# This document (Cognitive architecture & dataflow)
— manifest.jsonl	# Local cryptographic catalog of router & memory modules
— dual_agent_router/	# Triage vs. Executor handshake engine
— query_analyzer.py	# Evaluates user intent and checks token complexity
— executor_bridge.py	# Formats clean, compressed meta-prompts for the 9B model
— model_profile_registry.json	# Context window limits and quantization configs

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└─ mem0_episodic/          # In-session memory and dynamic preference caching
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| └─ vector_recall_engine.py  # Semantic lookup for active conversational facts
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| └─ rolling_habits.json     # User syntax preferences and operational habits
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| └─ ob1_static_protocols/   # Ground-truth static query bridges over MCP
|
└─ reasoning_bank/          # The Self-Healing System Flywheel
    └─ active_execution_paths.json  # Persistent JSON ledger tracking fractional task
    steps
        └─ failure_logs/      # Structured error reports for weekly pattern detection
        └─ post_regression/    # Rollback checkpoints to prevent capability decay
            └─ baseline_tests/  # Standardized test suites for on-device swarms
            └─ regression_audit_report.jsonl  # Historical accuracy metrics across model swaps

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Beginner-Proof Implementation Rules

1. **State Continuity:** If a model process panics or the phone restarts, the recovery engine reads `active_execution_paths.json` to resume the task at the exact step where execution halted.
2. **Zero Raw Chat Flooding:** `mem0` returns only condensed fact tokens—never dump raw multi-turn chat logs into the 9B model's prompt.